

Big Basket-E-commerce-Database-Management-System

Chapter 2

2.1 Introduction:

The entities required for the big Basket E-Commerce Database are identified based on grocery shopping requirements. The attributes of each entity along with their Primary Keys (PK) and Foreign Keys (FK) are defined to maintain proper data organization. The relationships between entities are analysed using cardinality and participation constraints. This design helps in creating the ER diagram, relational schema, and database implementation while ensuring data consistency and reducing redundancy.

2.2 Entity and their attributes and keys:

i. Entity: Customer(Strong Entity)

Attribute	Key Type
Customer_ID	Primary Key
Customer Name	Not Null
Email	Unique Key
Phone No	Unique Key
Date of Birth	-
Address	-

ii. Entity: Category(Strong Entity)

Attribute	Key Type
Category_ID	Primary Key
Category Name	Unique Key
Description	-

iii. Entity: Supplier(Strong Entity)

Attribute	Key Type
Supplier_ID	Primary Key
Supplier Name	Not Null
Contact_No	Unique Key
Address	-

iv. Entity: Product(Strong Entity)

Attribute	Key Type
Product_ID	Primary Key
Product Name	-
Category_ID	Foreign Key
Supplier_ID	Foreign Key
Brand	-
Price	Check (Price > 0)
Stock	Check (Stock >= 0)
Expiry_date	-

v. Entity: Cart(Strong Entity)

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_items	-
Total_amount	-

vi. Entity: Cart_Item(Weak Entity)

Attribute	Key Type
Cart_Item_ID	Primary Key
Cart_ID	Foreign Key

Product_ID	Foreign Key
Quantity	Check (Quantity > 0)

vii. Entity: Order(Strong Entity)

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_date	-
Order_status	Check ('Pending','Confirmed','Shipped','Delivered','Cancelled')
Total_amount	-

viii. Entity: Order_Item(Weak Entity)

Attribute	Key Type
Order_item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Price	-

ix. Entity: Payment(Strong Entity)

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_method	-
Payment_status	Check ('Pending','Success','Failed','Refunded')
Payment_date	-
Amount	-

x. Entity: Review(Strong Entity)

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check (Rating BETWEEN 1 AND 5)
Review_text	-
Review_date	-

xi. Entity: Delivery(Strong Entity)

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_address	-
Delivery_date	-
Delivery_status	Check ('Pending','Shipped','Out for Delivery','Delivered')

2.3 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial/Total
Customer	Place	Order	One-to- Many	Partial/Total
Customer	Makes	Payment	One-to-Many	Partial/Total
Customer	Writes	Review	One-to-Many	Partial/Total
Category	Categorizes	Product	One to Many	Total/Total
Supplier	Supplies	Product	One-to-Many	Partial/Total
Product	Include-in	Order_item	One-to-Many	Partial/Total
Order	Contains	Order_item	One-to-Many	Total / Total
Order	Generates	Payment	One-to-One	Total / Total

Order	Generates	Delivery	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial /Total
Cart	Contains	Cart_Item	One-to-Many	Total / Total
Product	Included-in	Cart_Item	One-to-Many	Partial / Total